

FORMALDEHYDE

Synonyms

Formalin, formic aldehyde, methaldehyde, methanal, methyl aldehyde, methylene oxide.

Description

Available as gas and aqueous solution (formalin). Commercial solutions contain 37-50% formaldehyde and up to 15% methanol. Used in insulating foams (urea-formaldehyde foam), urethane resins, plastics, rubber. Component of some pharmaceutical and cosmetic products. Formaldehyde-based resins are used as adhesives in plywood, particleboard and fiber board. Used as disinfectant, fumigant, and ingredient of embalming fluids. Formaldehyde vapours can result from burning wood, coal, gas and oil, and when formaldehyde-containing products decompose on exposure to moisture and heat. Formaldehyde is normal metabolite of body; adults may produce and metabolize > 50 g of endogenous formaldehyde daily.

Toxicity

Gas is water soluble and inhalation causes immediate irritation of upper respiratory tract. Ingestion of concentrated solutions (5-10%) can cause corrosive tissue injury. Ingestion also may lead to life-threatening metabolic acidosis. Sensitization can occur from dermal or inhalation (rare) exposures.

Mechanism of Toxicity

Formaldehyde precipitates proteins and causes coagulative necrosis similar to acids. Metabolic acidosis due to accumulation of formic acid (metabolite of formaldehyde).

Toxic Dose

Ingestion of 30-120 mL of a 37% solution has caused death in adults. An adult survived after ingesting 120 mL of a 10% solution. Vapour concentrations > 1 ppm can cause mucous membrane and ocular irritation. Solutions > 2% can cause skin irritation.

Case Reports

A 41-year-old ingested 120 mL of formaldehyde solution (37% formaldehyde, 12.5% methanol) and developed severe abdominal pain, cyanosis, hypotension, coma and renal failure. Metabolic acidosis (pH 6.87), elevated formic acid level and a blood methanol level of 13.4 mmol/L (43 mg/dL) were seen within 30 minutes of ingestion. Patient died 28 hours later.

A 41-year-old ingested 240 mL of a 37% formaldehyde solution and a 46-year-old ingested 120 mL of a 10% formaldehyde solution. Both patients developed corrosive gastritis and ulceration, and recovered following total or partial gastrectomy.

A 55-year-old ingested an unknown amount of formaldehyde solution containing no methanol and developed hypotension, respiratory insufficiency and metabolic acidosis. Gastroscopy revealed extensive congestion, diffuse necrosis and hemorrhage of gastric and duodenal mucosa. Patient developed acute renal failure requiring daily hemodialysis. Adult respiratory distress syndrome, sepsis, conjugated hyperbilirubinemia developed and patient died of cardiac failure 3 weeks post ingestion. Autopsy showed burns to the gastric mucosa, jejunum, ileum and part of colon, including sigmoid.

Pharmacokinetics

Rapidly absorbed from oral and inhalation exposure. Dermal absorption is minimal. Rapidly metabolized to formic acid and converted to carbon dioxide and water via folate dependent pathways. Formaldehyde is excreted in exhaled air as carbon dioxide or excreted in urine as formic acid and other breakdown products.

Clinical Effects

- **Topical:** Solutions > 2% are irritating; possible burns with more concentrated solutions. Contact or allergic dermatitis.
- **Ocular:** Irritation may occur after exposure to vapours. Splash contact with a 4% solution may cause immediate discomfort, optical irregularity and visual disturbances with complete resolution of symptoms within 24 hours. Permanent corneal opacification and loss of vision following contact with 40% solution; may be delayed 12 hours or more.
- **Inhalation:** Irritation to upper respiratory tract. Higher concentrations may produce cough, dyspnea, laryngeal edema, bronchitis, pneumonia and pulmonary edema. Asthma has been reported in hypersensitive individuals. Hepatotoxicity (animal data).
- **Ingestion:**
 - General:** Ingestion results in immediate onset of severe abdominal pain with nausea and vomiting (may be bloody). Possible corrosive gastritis with ulceration and necrosis. Systemic effects include metabolic acidosis, hypotension, tachycardia and dyspnea. Cardiovascular collapse, ARDS and renal failure with severe exposures.
 - HEENT:** Pharyngeal pain and hoarseness. Possible ulceration of mouth or throat.
 - CVS:** Hypotension, tachycardia; hypovolemic shock and cardiovascular collapse with large exposures.
 - Respiratory:** Shortness of breath, pleural effusions and infiltrates, ARDS.
 - Neurologic:** Headache, dizziness, lethargy, coma.
 - GI:** Immediate onset of nausea, vomiting (may be bloody), severe abdominal pain. Possible corrosive and hemorrhagic gastritis. Ulceration and necrosis of stomach with severe exposures; possible perforation.

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Esophageal (lower portion) or small bowel injury has been reported. Delayed onset stomach contracture and stricture or pyloric stenosis. **See also** ACIDS.

Hepatic: Hyperbilirubinemia (uncommon).

GU: Nephritis and acute renal failure.

Fluids/Lytes/Acid-Base: Anion gap metabolic acidosis due to accumulation of formic acid metabolite.

Blood: Hemolysis has been reported in a hemodialysis patient following exposure to formaldehyde-contaminated dialysate.

For large ingestions of products containing methanol, **see** METHANOL.

Treatment

1. **Topical:** Remove contaminated clothing. Wash skin thoroughly with soap and water. Treat burns as needed.
2. **Ocular:** Flush eyes for 15 minutes with a gentle stream of tepid water. Obtain ophthalmologic opinion.
3. **Inhalation:** Remove from exposure. Provide respiratory support as required. Administer oxygen as needed. In severe exposures, monitor blood gases, perform chest X-ray.
4. **Ingestion:** Do not induce vomiting. Dilute with a maximum of 120 mL ($\frac{1}{2}$ glass) milk or water for a child; a maximum of 240 mL (one glass) milk or water for an adult. Avoid larger volumes which may precipitate vomiting and aspiration. Do not give fluids to patients with difficulty breathing, severe abdominal pain, vomiting, or who are unable to swallow or protect their airway. After initial dilution, patients with significant ingestion should be kept NPO until after surgical or gastroenterologic consultation.
5. Role of nasogastric aspiration is controversial but may be considered in a recent (within 30 minutes), large ingestion to prevent further gastric injury. **See** ACIDS.
6. Activated charcoal is not recommended. Efficacy unknown and may impair visualization of burns during endoscopy.
7. Protect airway and assist ventilation as needed.
8. Maintain fluid and electrolyte balance. Treat metabolic acidosis with IV sodium bicarbonate and folic acid (or leucovorin) as required. **See also** FOLIC ACID/LEUCOVORIN antidote monograph.
9. Monitor vital signs, ECG, electrolytes, anion gap, blood gases, renal function, and methanol levels in cases where product contains methanol.
10. Perform chest X-ray and/or abdominal series as needed.

11. Hypotension unresponsive to IV fluids can be treated with vasopressors.
12. Endoscopy should be performed in symptomatic patients or intentional ingestions. **See** ACIDS.
13. Hemodialysis should be considered if acid-base disturbances are unresponsive to conventional measures.
14. Possible surgical intervention for corrosive gastritis or delayed pyloric stenosis. **See** ACIDS.
15. For products containing methanol, **see** METHANOL.

Key Points

- ✓ Causes coagulative necrosis similar to acids; extent of injury varies with concentration.
- ✓ Inhalation results in immediate irritation of upper respiratory tract.
- ✓ Ingestion produces immediate onset of severe abdominal pain with nausea and vomiting (may be bloody). Possible corrosive gastritis with ulceration and necrosis.
- ✓ Metabolic acidosis may result from formic acid metabolite.
- ✓ Cardiovascular collapse, ARDS and renal failure with severe exposures.
- ✓ For topical exposures, remove contaminated clothing, wash skin thoroughly with soap and water, treat burns.
- ✓ For inhalation, remove from exposure and provide respiratory support.
- ✓ For ingestions, dilute with small volume of fluid, then keep patient NPO. Patient may require GI or surgical consult. **See** ACIDS.
- ✓ Treat metabolic acidosis with IV sodium bicarbonate and folic acid (or leucovorin) as required. **See also** FOLIC ACID/LEUCOVORIN antidote monograph.